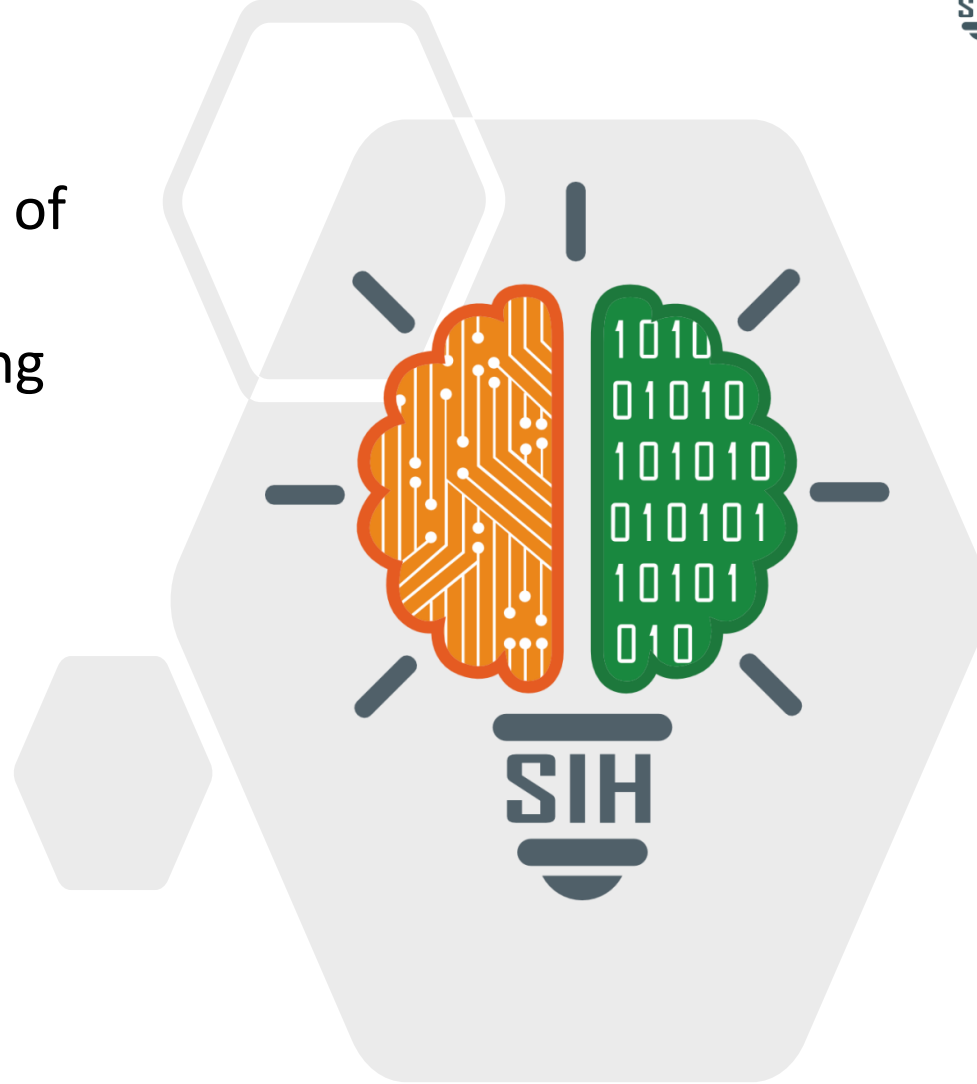


SMART INDIA HACKATHON 2026

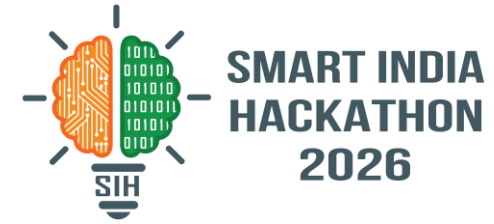


- **Problem Statement ID** – 26036
- **Problem Statement Title** – Development of an Online Verification System for Weighing and Measuring Instruments
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID-**
- **Team Name** - EvolveVI



Evolve
VI

Metro360 (Unified Digital Verification System for Weighing & Measuring Instruments)



ONE INSTRUMENT → ONE DIGITAL IDENTITY → COMPLETE LIFECYCLE

Digitizing the verification and certification of weighing & measuring instruments



PROBLEM



Verification & certification handled through physical forms, manual processes and office visits.



Leads to delays, duplicate records, missed re-verification deadlines and tracking difficulties.



Consumers do not have an easy way to verify the authenticity or track complaints.



IDEA / SOLUTION



Assign a unique Instrument Digital Identity (IDI) to every instrument.



Provide an end-to-end digital workflow from registration to re-verification.



Enable QR-based public verification + complaint portal for transparency and real-time tracking.



SOLUTION WORKFLOW



1. Register Instrument
(Manufacturer / Importer)

2. Verification & Certification
(LMO / GATC)

3. Create Digital Record
(IDI + Details)

4. QR for Public Access
(Verification + Complaints)



UNIQUE VALUE PROPOSITION



Unified ID per instrument across all stakeholders.



Eliminates paperwork and missed deadlines.



Instant verification and direct complaint channel for consumers.



KEY STAKEHOLDERS



Instrument Owners & Traders



LMOs



GATCs



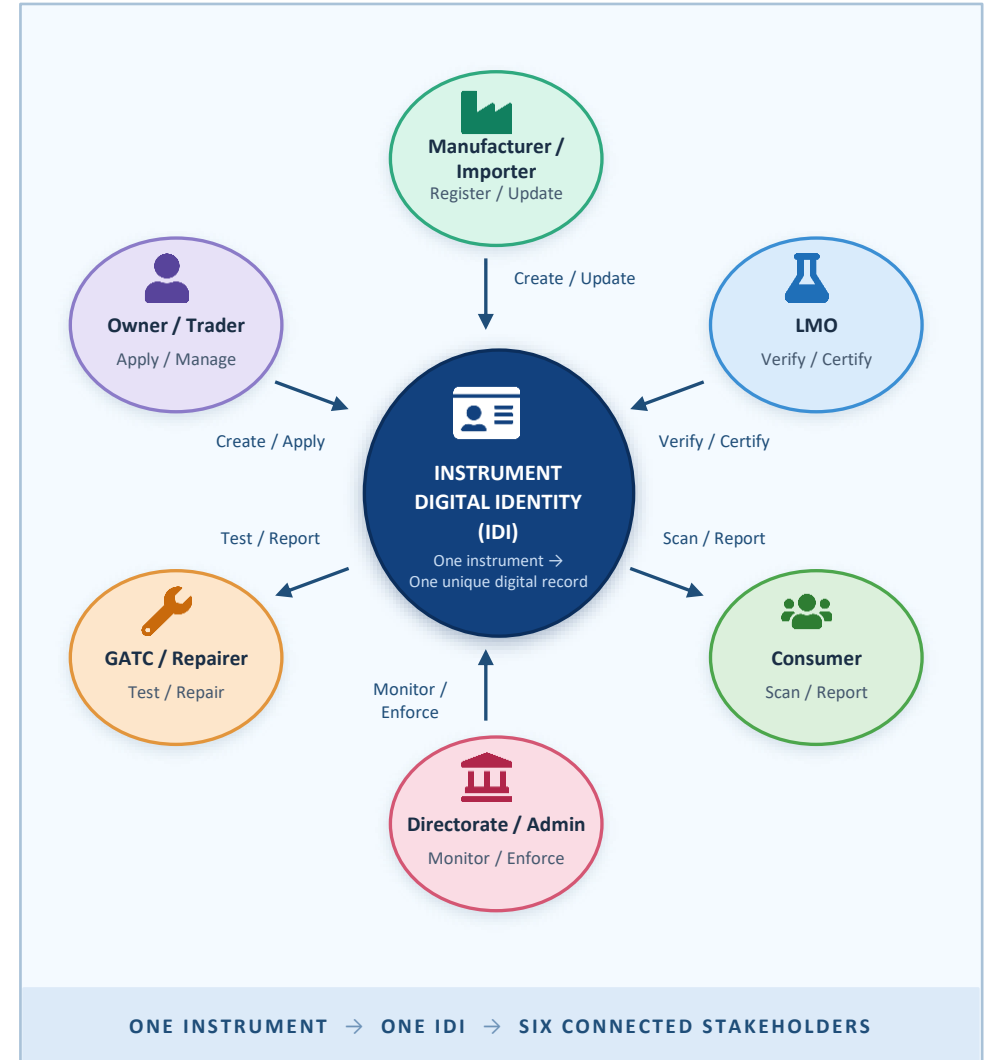
Manufacturers & Importers



System Admins & Metrology Directorate

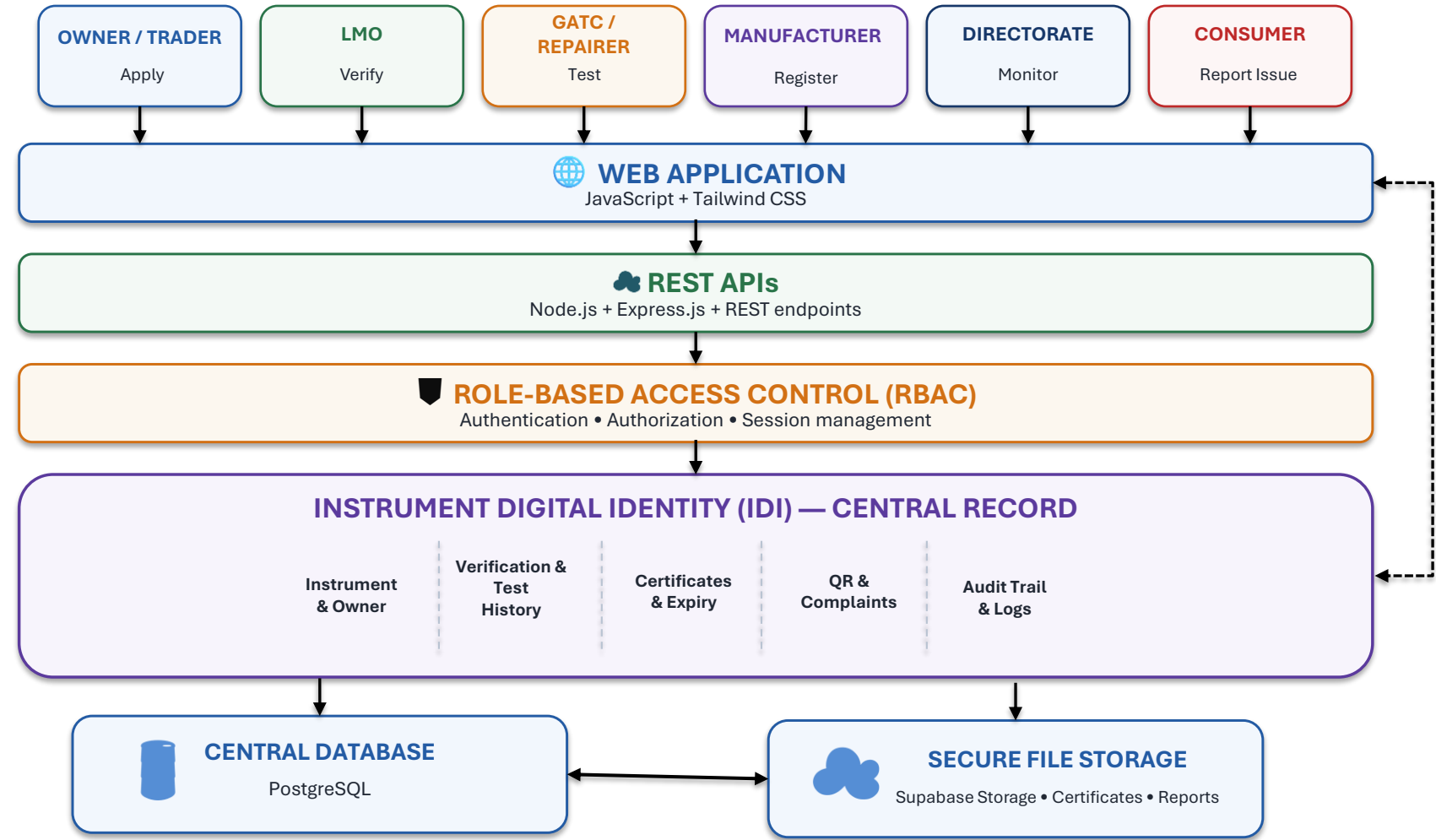


General Consumer



TECHNICAL APPROACH

STAKEHOLDERS / USERS



OPERATIONAL WORKFLOW



1) Technical Feasibility

- IDI + QR-based verification with centralized certificates and expiry records.
- Rule-based automated verification and compliance calculations.
- Modular web architecture designed for phased expansion.

2) Potential Challenges and Risks

- Maintaining accurate and up-to-date verification records.
- Preventing fake QR codes and unauthorized data changes.
- Integrating with diverse physical instruments and legacy systems in future phases.

3) Strategies for Overcoming Challenges

- Use Server-side certificate authentication and authenticated QR-linked records.
- Implement role-based access, audit trails and controlled record updates.
- Start with digital verification and certification; introduce hardware/IoT integration in future phases.

Impacts

Process Automation: Online applications, scheduling and certificate generation reduce paperwork and physical visits.

Centralized Tracking: A unified instrument repository provides traceable verification history across stakeholders.

Real-Time Enforcement: Dashboards help authorities monitor pending verifications, expiries and compliance actions.

QR-Enabled Authenticity: QR-based public verification enables quick authentication of certificates and instrument status.

Benefits

Higher Transparency:

Authenticated digital records reduce opportunities for unauthorized modification and improve traceability.

Time & Cost Savings:

Businesses avoid physical delays by applying and paying online. Speeds up application submissions, field data recording via mobile devices, and certificate retrieval.

Benefits to Stakeholders

Owner	Faster Application
LMO	Workload Visibility
GATC	Digital Test Reporting
Consumer	Verification + Complaint
Directorate	Enforcement Analytics
Manufacturer	Model/Bulk Registration

Better Consumer Protection:

Quick access to verified instrument status and certificate validity supports fair trade and consumer confidence.

S. No.	Title of the Paper / Documentation	Published / Presented in	Year of Publication	Key Insights
1	Legal Metrology Act, 2009 – Section 24	India Code / Government of India	2009	Establishes the legal requirement for verification and stamping of weights and measures used in transactions/protection and provides the legal foundation for the proposed digital verification workflow. (India Code)
2	Legal Metrology (General) Rules, 2011	Department of Consumer Affairs, Govt. of India	2011	Defines specifications and verification/re-verification requirements for weighing and measuring instruments, forming the rule base for digital verification and automated compliance calculations. (Consumer Affairs)
3	Legal Metrology (Government Approved Test Centre) Rules, 2013	Department of Consumer Affairs	2013	The official Department of Consumer Affairs page confirms the 2013 GATC Rules and their subsequent amendments. (Consumer Affairs)
4	Legal Metrology (Government Approved Test Centre) Amendment Rules, 2026	Department of Consumer Affairs, Government of India	2026	Updates the GATC framework and the scope of instruments eligible for verification through approved test centres, supporting digital GATC integration. (Consumer Affairs)
5	SIH 26036 – Online Verification System for Weighing & Measuring Instruments	Smart India Hackathon 2026 / Department of Consumer Affairs	2026	Identifies manual processes, isolated records and difficulty in tracking verification status; calls for centralized records, digital certificates, QR authentication, dashboards and mobile field verification. (SIH 2026 Problem Statements)
6	SIH 26035 – Test Report Generation for Non-Automatic Weighing	Smart India Hackathon 2026 / Department of Consumer Affairs	2026	Highlights the need for digital data entry, automated compliance calculations, standardized reports and instrument-wise history, supporting the digital verification ecosystem. (SIH 2026 Problem Statements)